

Pawan: Locally Calibrated Next-Day Air Quality Forecasting for Patiala

Statistical correction of a global forecast using station observations and upwind fire activity

Project Proposal for UCS503P

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1 Higher-order goal

... secondary framing

Every year between October and December, air quality across Punjab deteriorates sharply as paddy residue is cleared by burning. The wider problem, exposure to seasonal air pollution, is a matter of agricultural policy and public health, and no software project is going to solve it.

What software can do is smaller and more definite. If people knew a day in advance that tomorrow would be bad, some of what they do outdoors could be moved. Schools, sports coaching, construction supervision and morning exercise all involve decisions taken the previous evening. Those decisions are only as good as the forecast behind them.

The engineering objective, therefore, is to make next-day air quality information locally accurate for one city, to deliver it reliably without human intervention, and to be honest about how accurate it actually is.

2 Time-to-value

... primary heuristic

- The baseline against which we will be judged can be computed from public archives in week one. It needs no users, no interviews and no field study. We therefore know the “before” number before any development starts.
- Data collection is automated in week two. After that the training set grows on its own, and every later iteration begins with more data than the one before it, whether or not anyone touched the repository that week.
- Each iteration adds one feature or one modelling change, and reports the change in error on a fixed held-out period. Progress in a given week is therefore a number, not a claim.
- The first iteration succeeds if it beats a naive persistence forecast. The second succeeds if it beats the global model we are correcting. Neither milestone depends on anyone adopting the system.

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3 Problem Statement

Freely available air quality forecasts for Patiala come from global atmospheric models. The one that underlies most of them, the Copernicus Atmosphere Monitoring Service (CAMS) global model, runs at a resolution of about 0.4° , which is roughly 45 km per grid cell.

Patiala fits inside a single cell, with room to spare. So does the surrounding farmland, and a stretch of highway. The model treats all of it as one body of air and reports one number. When we queried the source at the station coordinates, the nearest grid point returned was 30.30°N , 76.40°E , and the value at that point is an average over an area far larger than the city.

Four consequences follow.

- **The city is invisible to the model.** Local emission sources, land use and terrain lie below the grid scale and cannot be represented at all.
- **The resulting error is systematic.** Because it arises from fixed local factors rather than from chance, it repeats. Anything that repeats can be learned and subtracted.
- **The error is worst when it matters most.** During the burning season, concentrations are driven by transient plumes carried from upwind fields. A coarse model resolves these poorly, so forecast quality degrades in precisely the weeks when a warning would be useful.
- **Nobody is checking.** No forecast published for Patiala is routinely scored against the city's own monitoring station, so a reader has no basis for deciding how much to trust it.

3.1 What we measured

Before proposing anything we scored the global forecast against the station's own record over 91 days, from 15 May to 14 August 2026. The result is worse than we expected.

- The observed daily mean was $24.6 \mu\text{g}/\text{m}^3$. The CAMS forecast for the same days averaged $3.13\times$ higher, a signed bias of $+52.3 \mu\text{g}/\text{m}^3$.
- Correlation between forecast and observation was $r = 0.38$, so roughly 15% of the variance in observed concentration is explained by the global forecast.
- Rescaling the forecast optimally, by fitting a straight line from forecast to observation on the same data, reduces its error to $8.5 \mu\text{g}/\text{m}^3$. That is still worse than assuming tomorrow resembles today, which achieves 6.5.

The conclusion shapes the whole project. The global forecast is not merely miscalibrated at this station, it is weakly informative. A correction that uses the forecast alone cannot succeed. What is needed is a local model in which recent observations carry the forecast, and the global model, meteorology and upwind fire activity contribute additional information at the margin.

Why this problem, in this place, now. Patiala has a continuous ambient air quality monitoring station at Model Town, operated by the Punjab Pollution Control Board, whose hourly readings are published nationally. Ground truth is already available and costs nothing. Equally relevant, the project window runs from August to December, which contains the burning season in full. The system will be built before the event, deployed during it, and evaluated against it. That alignment is unusually favourable and is a large part of why we chose this problem.

4 Proposed Solution

... *Software Proposition*

4.1 Overview

We propose a **web-based** service that forecasts tomorrow's concentration at one station, using every signal available on the previous evening. Recent observations at the station form the backbone, since they are the strongest single predictor. The global forecast, the local meteorological forecast and upwind fire activity are additional inputs, each retained only if it demonstrably reduces error.

We do not attempt to replace the global model. Replacing it would be a research project, and a hopeless one at this scale. Nor do we treat it as authoritative, because our own measurements show it is not. We treat it as one input among several, and we say by how much it helps.

- **Automated ingestion.** A scheduled job collects the previous day's observed concentrations, the forecast issued for the coming day, the local meteorological forecast, and satellite fire detections over the upwind region.
- **Feature assembly.** A reproducible transformation from raw records to a modelling table, with the forecast issue time carried on every row.
- **Correction model.** A supervised regression that predicts tomorrow's mean concentration from the global forecast plus local context.
- **Evaluation harness.** A rolling-origin backtest that scores the model and every baseline over the same periods, so comparisons are like for like.
- **Public dashboard.** One page: tomorrow's corrected forecast beside the uncorrected one, recent observations, current accuracy, and the timestamp of the last successful pipeline run.
- **Advisory layer.** A rule converting the predicted concentration into an air quality band and a short scheduling recommendation.

4.2 Core workflow

1. The scheduled job runs each morning and appends one record: yesterday's observation, the forecast issued for the coming day, the meteorological forecast, and the upwind fire count.
2. The feature builder assembles a modelling table from the accumulated records, admitting only quantities that existed at forecast issue time.
3. The model is fitted on the historical portion of that table and emits a corrected value for the next day.
4. The backtest scores model and baselines on a held-out period and writes the metrics to a file.
5. The dashboard is rebuilt from the prediction and metric files, so what is published always matches the last successful run.

4.3 Operating constraints we have imposed

- Every data source must be free and publicly reachable, with no institutional permission required. We verified this experimentally before writing this proposal rather than assuming it.
- The system must run unattended. Any step that needs a person on a given morning is a defect, not a workflow.

- No new algorithm is to be introduced. Statistical correction of numerical model output is established operational practice. We are applying it, not extending it.
- The page must be usable on a mid-range phone on a slow connection, because that is how anyone would actually check it.

5 Solution Approach

... *Engineering focus*

This is an engineering course project. The substance lies in whether the data pipeline is correct, whether the evaluation is honest, and whether delivery is reliable. It does not lie in the choice of estimator.

5.1 Correction method

... *non-research*

Post-processing numerical model output using local observations is a long-established technique, known in operational meteorology as model output statistics. Our approach is deliberately ordinary:

- Predict tomorrow's mean concentration from the issued global forecast, recent observations, forecast meteorology, calendar features and upwind fire activity.
- Fit a gradient boosted regression tree ensemble, a standard choice for small tabular datasets, using an established open source library.
- Fit an ordinary linear model first and report it alongside, so that any benefit from the more complex model is demonstrated rather than presumed. If the linear model turns out to be close, we will say so.
- Claim nothing about the state of the art. The claim is narrow and checkable: error at this station falls by a stated margin relative to the uncorrected forecast.

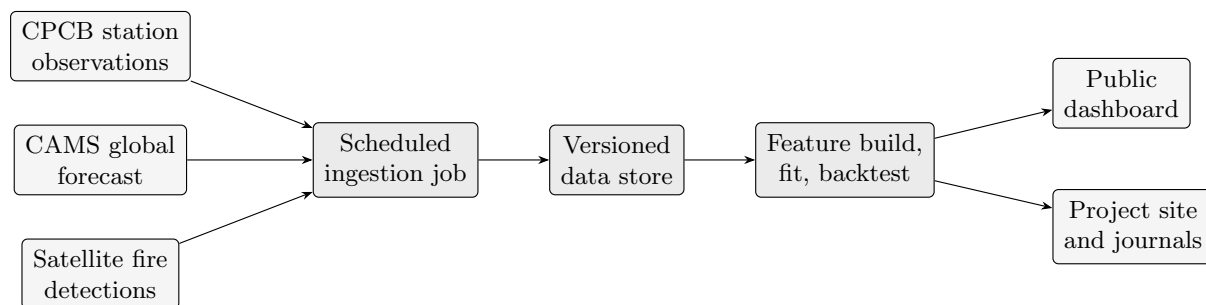
5.2 Two failure modes designed against

Both of the following would produce excellent numbers and a useless system, which is what makes them dangerous.

- **Temporal leakage.** The forecast that is *valid* for a given day is a different object from the forecast that was *issued* the day before for that day. Only the second exists when a prediction has to be made. The ingestion job therefore records issue time on every row, and the feature builder rejects any input timestamped after the issue time of the prediction it feeds.
- **Improper splitting.** A random train-test split is invalid for time series, because it permits fitting on data that follows the evaluation window. Evaluation uses a rolling-origin backtest instead, refitting on data preceding each simulated forecast date.

5.3 System architecture

... web-based solution



The design is file-based rather than service-based. Ingested records and computed metrics are stored as versioned files in the project repository. This has a useful side effect: every historical state of the dataset, and every forecast we ever published, remains auditable afterwards. There is no server to operate and no database to administer for the duration of the course.

5.4 CI/CD as project infrastructure

Continuous integration and deployment are in place before the first model is fitted, because the central artefact of this project is a job that must run by itself every day.

- A scheduled workflow performs daily ingestion without human involvement and records whether it succeeded. From week two onward there is a running production system, however small.
- A workflow on every proposed change runs the test suite, covering parser behaviour and schema validation of ingested records.
- Merging an accepted change rebuilds and redeploys both the dashboard and the documentation site, so published artefacts cannot drift away from the repository.
- Credentials live as encrypted repository secrets and are injected at run time. They do not appear in source at any point.

6 Evaluation Criterion

... measurable and attributable

Every metric below is computed from timestamped machine records. None depends on a survey, on self-report, or on recruiting anyone.

6.1 Primary metric

Mean absolute error of the next-day mean PM_{2.5} concentration at the Model Town, Patiala station, in $\mu\text{g}/\text{m}^3$, over a held-out period, reported as the percentage reduction against the *persistence* baseline.

Persistence is the primary bar rather than the global forecast, and the choice is deliberate. Our measurements show the global forecast is so poorly calibrated here that beating it is not an achievement: a two-parameter rescale reduces its error by roughly 84% on the first attempt. Persistence, by contrast, achieves $6.5 \mu\text{g}/\text{m}^3$, which is 26% of the observed mean, and is genuinely difficult to improve upon. Setting the easier reference as the headline would misrepresent the work.

- **Target, low season:** a reduction of at least 10% in MAE relative to persistence, over a held-out period drawn from the monsoon and post-monsoon regime.

- **Target, burning season:** a reduction of at least 20% relative to persistence during October and November. We expect a larger margin here, because day-to-day variability rises sharply when transported smoke dominates, and persistence degrades fastest in exactly that regime. This is the evaluation that matters.
- **Attribution:** model and baselines are scored against the same observation series over the same days, so differences between them are attributable to the model and to nothing else.

6.2 Measured baselines

The following were computed from public archives over 91 days, 15 May to 14 August 2026, before any development began. They fix the “before” condition of the project.

Reference	Definition	MAE ($\mu\text{g}/\text{m}^3$)
Observed mean	mean of observed daily values	24.6 [†]
Persistence	tomorrow equals today	6.5
Climatology	mean of the observed period	9.8
Rescaled CAMS	global forecast, linearly refitted	8.5
Raw CAMS	global forecast, uncorrected	52.3

[†] a level, not an error; included so the errors above can be read in proportion.

Three observations follow, and each shaped a design decision.

- Raw CAMS error exceeds the observed mean itself by a factor of more than two. The forecast is unusable at this station without correction, which establishes that the problem is real.
- Even optimally rescaled, the global forecast remains worse than persistence. Correction of the forecast alone is therefore not a viable design, and the model is built around local observations instead.
- These figures describe the monsoon and post-monsoon period, when concentrations are low and stable. They are the easy regime. Both baselines will deteriorate substantially once burning begins, and we will report the seasonal comparison rather than a single annual figure.

6.3 Secondary metrics

- **Category recall.** Of the days on which observed air quality reached the *Very Poor* band or worse, the fraction flagged twenty-four hours ahead. The advisory stands or falls on this number.
- **False alarm rate.** The fraction of issued advisories for days that did not reach the flagged band. Reported next to recall so the trade-off between the two is visible rather than hidden.
- **Pipeline success rate.** The proportion of scheduled days on which a forecast was published without anyone intervening. This measures the engineering, not the model.
- **Data availability.** The proportion of expected hourly observations actually received. Monitoring stations go offline; we would rather report that than quietly drop the rows.

6.4 Validation plan

... *high-level*

- Baselines are computed in the first iteration from archived data, fixing the “before” condition ahead of development.
- Every iteration is scored on the same held-out period, so improvements are comparable across weeks rather than only within one.
- The final evaluation covers the burning season in full. That is the regime the system exists for and the hardest test available inside the project window.

7 Scalability

... *with foresight*

1. **Scalable in principle.** The design generalises along three independent axes without structural change.
 - *Spatial.* The monitoring station is a configuration value, never a constant in code. Additional stations are additional configuration entries and additional fits, since the correction is per-station by construction.
 - *Pollutant.* The target variable is a parameter. Applying the same method to PM₁₀ or ozone changes no ingestion or evaluation code.
 - *Horizon.* The upstream forecast extends to five days. Extending the corrected horizon changes the target column and nothing else.
2. **Deployable in practice.** The substitutions needed to turn this into a maintained service are known now, and confined to storage and serving.
 - Versioned files become a managed relational database with the same schema.
 - The scheduled repository workflow becomes a hosted scheduler.
 - The static page becomes a small read-only API returning the same shape.
 - Modelling, feature and evaluation code are untouched by all three.

8 Engine availability heuristic

Everything required is mature, documented and free at the volume we need. We tested each source before writing this proposal, and record the outcome here rather than asserting availability in the abstract.

- **Ground truth.** The Central Pollution Control Board real-time feed, published through the national open data platform. Verified to return current PM_{2.5} readings for the Model Town, Patiala station.
- **Forecast source.** An open API serving the CAMS global model. Verified to return a complete ninety-two day hourly archive for the Patiala coordinates with no missing values. Its accuracy at this station was then measured rather than assumed, with the results given in Section 6.
- **Fire activity.** NASA’s Fire Information for Resource Management System, serving satellite detections by bounding box. Verified reachable with a valid response structure. One caveat surfaced during testing and is recorded under risks.
- **Historical observations.** The Central Pollution Control Board station archive, used once to establish the initial training and baseline period.

- **Modelling.** Established open source Python libraries for tabular data handling and gradient boosted regression.
- **Automation and hosting.** A hosted continuous integration runner for scheduled and triggered workflows; static hosting for the dashboard and documentation.
- **Testing.** A standard Python test framework, applied to parsers, schema validation and evaluation code.

9 Project scope and deliverables

... iteration-friendly

9.1 First iteration

- Ingestion adapters for the observation feed, the forecast source and the fire detection service.
- A scheduled workflow appending one validated record per day, carrying forecast issue time.
- Baselines computed for persistence, climatology and the uncorrected forecast over a fixed period.
- A first correction model shown to beat persistence.
- Continuous integration running the test suite on every proposed change.
- A published dashboard showing corrected and uncorrected forecasts for the next day, recent observations, current error statistics, and last successful run time.

9.2 Later iterations

- Upwind fire activity and wind direction features, introduced ahead of the burning season.
- Air quality band classification and the advisory rule, with recall and false alarm rate reported together.
- A rolling-origin backtest covering the full season.
- Scheduled retraining, with the interval chosen from measurement rather than assumed.
- Feature importance analysis, used to justify keeping or dropping each input.

9.3 Explicitly excluded

Naming the exclusions is part of defining the scope, and keeps the deliverable bounded.

- Medical or clinical recommendations of any kind. Output is limited to scheduling guidance for outdoor activity.
- User accounts, authentication, or personal data. The service stores nothing about anyone.
- Multiple cities within the course window, even though the design admits them.
- Any modification of the underlying atmospheric model, or any claim about atmospheric processes beyond the statistical relationship we measure.

10 Risks and mitigations

- **Station outage.** Continuous monitoring stations drop out, sometimes for weeks. The station is a configuration value, so the project can move to another Punjab station without code change. Availability is reported as a result rather than concealed.
- **Distribution shift at season onset.** A model fitted on monsoon and post-monsoon air will meet a different regime when burning starts. We address this with scheduled retraining, and intend to report the resulting error trajectory as a finding rather than smooth it away.
- **Temporal leakage.** The most serious correctness risk, and invisible from performance figures alone. Mitigated by recording issue time from the first ingestion run and by rolling-origin evaluation.
- **Archive retention limits.** Testing revealed that the fire detection service serves near-real-time and science-quality products under different names, with different retention windows, and returns an empty result rather than an error when a requested date falls outside the window. Source selection is therefore driven by the availability endpoint rather than hard-coded.
- **Silent interface failure.** Public interfaces change without notice, and some fail quietly. During development we found that the observation API discards unrecognised query parameters rather than rejecting them, so a mistyped date filter returned the entire series and looked like a successful narrow query. Each source therefore sits behind a single adapter that validates the *content* of a response, not only its status code, and every filtered query is re-checked client side.
- **Ambiguous station identity.** The monitoring station is exposed through two sensor identifiers for the same pollutant, one of which stopped reporting in 2022 and still returns a plausible historical series. The active identifier is pinned in configuration with the reason recorded, so the choice cannot be silently reversed by someone reading the source listing later.
- **Shallow archive of issued forecasts.** The accessible archive of issued forecasts is bounded, which limits model complexity. We address this by preferring low-variance models with few parameters, and by beginning automated accumulation immediately.
- **Team bandwidth.** Ingestion, modelling and delivery each have one responsible member, and the pipeline continues collecting data through examination weeks without attention.

11 Summary

We propose a service that corrects a coarse global air quality forecast at one monitoring station in Patiala, and publishes the corrected next-day value with a short advisory.

Against the course criteria: time-to-value is high, because the evaluation baseline comes from public archives before development starts and the pipeline becomes self-sustaining in week two. The evaluation criterion is measurable and attributable, because corrected and uncorrected forecasts are scored against the same observations over the same days. CI/CD is structural rather than decorative, because the central artefact is a job that must run unattended every morning. The engines are mature, free and verified. The proposition scales along spatial, pollutant and horizon axes without redesign.

Finally, it is relevant where it is built. The system addresses a pollution event that occurs in this region during this semester, and is evaluated against this region's own monitoring data.